



PES University, Bengaluru
(Established under Karnataka Act 16 of 2013)

END SEMESTER ASSESSMENT (ESA) - Jan-Feb

UE24PH151A - Engineering Physics

Total Marks : 100.0

1.a. Write the expression for $E(z, t)$ and $B(z, t)$ of electromagnetic waves. Using Maxwell's equations, show that electric and magnetic vectors are orthogonal? (5.0 Marks)

1.b. Write any two different forms of Heisenberg Uncertainty Principle? Calculate the momentum of an electron and the de-Broglie wavelength associated with it if its kinetic energy is 1.5 keV. (Given, mass of the electron $m = 9.1 \times 10^{-31}$ kg, Charge of the electron $e = 1.6 \times 10^{-19}$ C, Planck's constant $h = 6.626 \times 10^{-34}$ Js) (6.0 Marks)

1.c. Distinguish between phase and group velocity? Show that the de-Broglie wave group associated with a moving particle travels same velocity as that of particle? (5.0 Marks)

1.d. What is an operator in quantum mechanics? Starting with the wave function $\psi = Ae^{-i(\omega t - kx)}$ obtain the operator for kinetic energy and momentum? (5.0 Marks)

1.e. Explain the following terms,
i. Compton shift
ii. Observables (4.0 Marks)

2.a. Solve the Schrodinger equation for the particle ($V_0 < E$) incident on a step potential of height V_0 and show that $T+R=1$ (8.0 Marks)

2.b. Set up the one-dimensional time independent Schrodinger equation. (5.0 Marks)

2.c.
Write the normalized wave functions and sketch the Eigen functions and probability density curves of the first three allowed quantum states for a particle in an infinite potential well.

(4.0 Marks)

2.d. Calculate the probability of transmission for an electron of energy 1 eV incident upon a rectangular potential barrier of height 10 eV and width 0.50 nm. (Given, mass of the electron $m = 9.1 \times 10^{-31}$ kg, Charge of the electron $e = 1.6 \times 10^{-19}$ C, Planck's constant $h = 6.626 \times 10^{-34}$ Js) (4.0 Marks)

2.e. How do the potential and Eigenvalues of a particle exhibiting simple harmonic oscillations differ from that of a particle trapped in an infinite potential well?

The ground state energy of an electron in an infinite potential well is 5.6×10^{-3} eV. What will be the ground state energy if the width of the potential well is doubled? (Given, mass of the electron $m = 9.1 \times 10^{-31}$ kg, Charge of the electron $e = 1.6 \times 10^{-19}$ C) (4.0 Marks)

3.a. What is the significance of fermi factor? Graphically represents the variation of fermi factors with energy for $T = 0K$ and $T > 0K$.

Find the relaxation time of conduction electrons in a metal of resistivity $1.54 \times 10^{-8} \Omega m$, if metal has 5.8×10^{28} conduction electron per m^3 . (Given, mass of the electron $m = 9.1 \times 10^{-31}$ kg, Charge of the electron $e = 1.6 \times 10^{-19}$ C) (6.0 Marks)

3.b. Mention the failures of Quantum free electron theory?

Elaborate with suitable schematic the dependence of effective mass of electrons on the curvature of the E-k plot? (4.0 Marks)

3.c. Explain the terms, i. Drift velocity, ii. Critical field, iii. Bohr Magneton.
(6.0 Marks)

3.d. Discuss the temperature dependent susceptibility for different types of magnetic material with neat diagram?
(4.0 Marks)

3.e. Briefly describe the functionality of a GMR device and mention its Important applications.
(5.0 Marks)

4.a. Derive an expression for energy density in terms of Einstein coefficients? (6.0 Marks)

4.b. Discuss the requirements of a laser system?
Calculate the threshold gain factor of a laser system, which has a loss factor of 0.05 m^{-1} , with tube length of 50 cm having one mirror 99% reflecting and the output coupler 90% reflecting. (5.0 Marks)

4.c. Discuss the construction and working of homojunction semiconductor laser with neat diagram? (6.0 Marks)

4.d. Compare the phenomenon of ferroelectricity and piezoelectricity.

(4.0 Marks)

4.e. Discuss the dielectric polarization dependence on the electric field in a linear dielectric.

(4.0 Marks)



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1.a. Define the following with respect to electromagnetic fields

(i) Displacement current

(ii) Poynting vector (magnitude and direction)

(4.0 Marks)

1.b. (i) Write the equations of conservation of momentum and conservation of energy in Compton's scattering of photons by electrons.

(ii) A photon with a wavelength of 0.8 nm undergoes two successive Compton scattering events. The first scattering occurs at an angle of 60° , and the second scattering happens at 120° . Determine the total energy lost by the photon. **(6.0 Marks)**

1.c. A photon's position is measured with an uncertainty **10 nm** .

(i) Use the Heisenberg Uncertainty Principle to calculate the minimum uncertainty in the photon's momentum.

(ii) Estimate the uncertainty in the photon's energy. **(4.0 Marks)**

1.d. (i) Discuss the properties of a well behaved quantum wavefunction.

(ii) Consider the following wavefunction for a particle in one dimension:

$$\psi(x) = \begin{cases} Ax, & 0 \leq x \leq L \\ 0, & \text{otherwise} \end{cases}$$

Determine the normalization constant.

(6.0 Marks)

1.e. (i) Using suitable operators, set up the Schrodinger's time independent equation.

(ii) Write the solutions to the Schrodinger's time independent equation for a free particle.

(5.0 Marks)

2.a. Consider a particle in a step potential $V(x) = \begin{cases} 0, & x < 0 \\ V_0, & x \geq 0 \end{cases}$

(i) Write the Schrodinger's time independent equation and solutions in both the regions for a case $E < V_0$.

(ii) Interpret the significance of the wavefunction in the region $x \geq 0$.

(4.0 Marks)

2.b. Explain the concept of quantum tunneling. On what factors does the tunneling probability depend? (4.0 Marks)

2.c. (i) Obtain the Eigen values and normalized Eigen functions for particle in a 1D infinite well with boundaries at $-\frac{a}{2}$ & $\frac{a}{2}$

..
(ii) Determine the expectation value of energy for a particle in 1D infinite well. (7.0 Marks)

2.d. What is anharmonicity of a 'quantum harmonic oscillator' and how are practical qubit systems prepared for quantum computation? Give an example of a practical qubit system. (4.0 Marks)

2.e. Obtain an expression for the Density of States for a quantum free electron gas model and show its variation with energy. (6.0 Marks)

3.a. Discuss any two demerits of Drude's classical free electron gas model. (4.0 Marks)

3.b. Define the following

- (i) Fermi energy
- (ii) Fermi speed
- (iii) Fermi temperature
- (iv) Effective mass

(4.0 Marks)

3.c. Define the following with mathematical expressions and plots

(i) Periodic potential 1D

(ii) Bloch function for 1D

(iii) E-k diagram (6.0 Marks)

3.d. (i) Why does an atomic magnetic dipole precess about a magnetic field ? Calculate the Larmor frequency of an electron in an atom when a magnetic field of **4 T** is applied.

(ii) What is Lande's g factor? Calculate its value for an electron with pure spin moment.

(6.0 Marks)

3.e. Discuss how the susceptibility of ferromagnetic and antiferromagnetic materials varies with temperature along with a mention of important properties.

(5.0 Marks)

4.a. Discuss the concepts of absorption, spontaneous emission and stimulated emission along with the rate equations. Why cannot spontaneous emission produce a LASER?

(6.0 Marks)

4.b. (i) What are the roles of a resonant cavity?

(ii) Derive an expression for the round trip Threshold Gain Coefficient.

(4.0 Marks)

4.c. (i) Discuss lasing in He-Ne laser with an energy level diagram.

(ii) A LASER has a wavelength of $1064 \text{ nm} \pm 2 \text{ nm}$. Find the coherence time.

(6.0 Marks)

4.d. (i) How does the electric Polarization depend on the Electric field? Calculate the polarization if the dielectric constant of a sample is 80 and the net electric field is 44 Vm^{-1} .

(ii) List the polarization mechanisms in dielectrics. (4.0 Marks)

4.e. (i) Discuss the structure and properties of Barium Titanate.

(ii) A laser beam of wavelength 1064 nm passes through a second order nonlinear optical crystal. Determine the frequency of the converted beam. (5.0 Marks)



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END SEMESTER ASSESSMENT (ESA) - JAN 2023

UE23PH151A - Engineering Physics

Total Marks : 100.0

1.a. Write the Maxwell's equations in differential form (for free space). (4.0 Marks)

1.b. What is the UV catastrophe in the Rayleigh model of black body radiation? Explain how this was overcome in Planck's model of black body radiation. (4.0 Marks)

1.c.
Find the expectation value of position "x" of a particle trapped between $x=0$ to $x=L$ in a state, $\varphi(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right)$. (5.0 Marks)

1.d. Define the following terms:

(i) Wave function (ii) Operator (iii) Expectation value (iv) Probability flux

(4.0 Marks)

1.e.

Set up the one dimensional time independent Schrodinger equation for a non-relativistic particle of mass 'm' from time dependent Schrodinger wave equation.

(8.0 Marks)

2.a. Obtain an expression for probability of a particle of mass "m" and energy "E" being reflected back when incident on a potential step of height " V_0 " (such that $V_0 < E$) using appropriate boundary conditions.

(7.0 Marks)

2.b.

The tip of Scanning Tunnelling microscope is 0.18 nm away from the metal surface. The metal-tip gap is a potential barrier of height $U = 12$ eV. What is the probability that an electron of energy $E = 6$ eV in the metal will tunnel through the barrier? Given: $\hbar = 6.626 \times 10^{-34}$ J.s, $m = 9.1 \times 10^{-31}$ kg

(5.0 Marks)

2.c. Write an expression for potential energy and energy eigenvalues of hydrogen atom and 1d harmonic oscillator. Additionally sketch the wave function of first excited state of 1d harmonic oscillator.

(5.0 Marks)

2.d.

Obtain an expression for Fermi energy of an electron in a metal using the concept of density of states.

(6.0 Marks)

2.e.

Find the Fermi energy of Potassium given its Fermi temperature is 24600K.

(Given: $k_B = 1.38 \times 10^{-23} \text{ J/K}$)

(2.0 Marks)

3.a.

A uniform silver wire has a resistivity of $1.54 \times 10^{-8} \Omega\text{-m}$ at room temperature.

For an electric field of 1 V/cm, calculate (i) Mobility (ii) Drift velocity.

Given: $n = 5.82 \times 10^{28}$ conduction electrons per m^3 , $e = 1.6 \times 10^{-19} \text{ C}$.

(4.0 Marks)

Define the following terms:

3.b. (i) Fermions (ii) Effective mass (iii) Bloch function (iv) Energy gap
(v) Hall effect

(5.0 Marks)

How did Quantum free electron theory successfully explain the following?

- 3.c. (i) Dependence of electrical conductivity of metals on temperature
(ii) Electronic contribution to specific heat

(4.0 Marks)

3.d.

What is the magnetic interaction energy of a magnetic dipole in an external field? Arrive at an expression for magnetisation of paramagnetic materials when placed in an external magnetic field such that the thermal energy dominates magnetic interaction energy?

(7.0 Marks)

- 3.e. What are Superconductors? Classify them based on how they respond to the external magnetic field with an example each.

(5.0 Marks)

4.a.

What are the three ways in which matter interacts with radiation? Mention the factors on which these processes rates depend on.

(6.0 Marks)

4.b.

With an appropriate energy level diagram explain how laser transition occurs in He-Ne laser.

(5.0 Marks)

4.c.

Laser light is highly monochromatic but never perfectly monochromatic. Substantiate the statement with appropriate reasons for it being highly monochromatic as well as for being never perfectly monochromatic.

(4.0 Marks)

4.d.

The dielectric constant of helium atom is 1.0000684. If the Helium gas contains 2.7×10^{26} atoms/m³, calculate (i) Electronic polarizability (ii) Radius of helium atom.

Given: $\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$

(5.0 Marks)

4.e.

Write a note on piezoelectric and pyro electric materials including their behaviour in the presence of external electric field, one example and application each.

(5.0 Marks)



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END SEMESTER ASSESSMENT (ESA) - JAN 2023

UE23PH151A - Engineering Physics

Total Marks : 100.0

1.a. Using Maxwell's equations for free space, obtain the equation of wave propagation describing the magnetic field. (5.0 Marks)

1.b. Explain the uncertainty principle. How does the analysis of gamma ray microscope thought experiment establish the uncertainty principle. (5.0 Marks)

1.c. Derive Schrodinger's time independent wave equation for a particle. (6.0 Marks)

1.d. Explain group velocity and phase velocity. Derive the relation between group and phase velocities. (5.0 Marks)

1.e. X-rays of wavelength 0.113 nm is scattered from a carbon target. Calculate:
(i) the wavelength of X-rays scattered at an angle of 90° with respect to the original direction.
(ii) the energy acquired by the scattered electron. (4.0 Marks)

2.a. A particle of mass 'm' and total energy 'E' moves towards a one dimensional potential step of height V_0 ($E > V_0$), by applying continuity conditions obtain the expression for transmission coefficient. (6.0 Marks)

2.b. A beam of electron of energy 1.9 eV approaches a potential step of height 4.8 eV. To what depth it can penetrate into the classically forbidden region? (3.0 Marks)

2.c. Discuss how in radio activity, 'alpha particle emission' is explained as a case of barrier tunneling. (4.0 Marks)

2.d. Write and explain Fermi-Dirac distribution function.
Show that:
(i) at all temperatures ($T > 0K$) probability of occupancy of Fermi level is 50%.
(ii) the probability that a state ΔE above the Fermi level E_F is filled equals the probability that a state ΔE below is empty. (6.0 Marks)

2.e. Discuss the probability densities and eigen energy values for a particle in a one dimensional potential well of infinite height considering the first two excited states.
Calculate the energy of the electron in the energy level immediately after the lowest energy level confined in a cubical box of 0.12 nm side.
(6.0 Marks)

3.a. Explain ferromagnetism, antiferromagnetism and ferrimagnetism on the basis of magnetic dipoles of the atoms.
(6.0 Marks)

3.b. What is Meissner effect? Explain briefly the salient points of BCS theory.
(5.0 Marks)

3.c. Calculate the mobility and the relaxation time of electrons in monovalent copper with the following data:

Resistivity of copper = 1.73×10^{-8} ohm-m; atomic weight = 63.5; density = 8.92 g/cm^3 , Avogadro's number = 6.02×10^{23} (4.0 Marks)

3.d. Show that the Lorenz number obtained as per Wiedemann-Franz law is equal to $\frac{\pi^2}{3e^2} k_B^2$. (5.0 Marks)

3.e. Explain briefly:
(i) Bloch function.
(ii) Effective mass and its dependency on curvature of E-k plot. (5.0 Marks)

4.a. Briefly explain holography. With a neat sketch explain the off-axis holographic recording process. (5.0 Marks)

4.b. With suitable schematic sketch, explain the construction and working of a semiconductor laser. List the features that are responsible for the higher efficiency of double heterojunction devices. (6.0 Marks)

4.c. The lifetime of transitions in a system emitting wavelength of 620 nm is estimated to be 19 ns.
Calculate:
(i) Einstein's coefficients A and B.
(ii) spectral broadening and coherence length of radiation. (4.0 Marks)

4.d. Derive a relation between polarization of a dielectric and the external electric field. (4.0 Marks)

4.e. Distinguish between piezoelectricity and pyroelectricity. Sketch the property response of a ferroelectric nonlinear dielectric in an external field. (6.0 Marks)



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END SEMESTER ASSESSMENT (ESA) - Feb 2026

UE25PH151A - Engineering Physics

Total Marks : 100.0

1.a. The electric field component of an EM wave obeys the wave equation $\frac{\partial^2 E}{\partial t^2} = \frac{1}{\mu_0 \epsilon_0} \frac{\partial^2 E}{\partial x^2}$. If the solution to this wave equation is $E = E_0 \sin(kx - \omega t)$ then obtain a relation connecting k , ω , μ_0 and ϵ_0 . (5.0 Marks)

1.b. What are phase and group velocities? Derive a general relation between the phase and group velocities involving the wavelength of the waves. (5.0 Marks)

1.c. According to de Broglie what is the relation between the momentum of a particle and the wavelength of the wave associated with it? Two metal plates are placed apart in a vacuum. The potential on one plate is +50V and the other -50V. An electron is released from the plate with negative potential, with zero initial velocity, and is received by the plate with positive potential. Estimate the de Broglie wavelength of the electron before it is captured by the plate with positive potential. (5.0 Marks)

1.d. Explain the term normalization (assume the system to 1D) . A Gaussian wavepacket, centered about the point x_0 is given by

$\psi = Ae^{-\frac{(x-x_0)^2}{4\sigma^2}}$, where σ is a constant. The limits of the wavepacket is defined by $-\infty \leq x \leq \infty$. Estimate the value of A, the amplitude of the wavefunction. You are given the relation $\int_{-\infty}^{\infty} e^{-y^2} dy = \sqrt{\pi}$. Here y is some variable.

(5.0 Marks)

1.e. The state of a quantum system is given by $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. Answer the following

(i) What does the relation represent?

(ii) What is the notation $|\psi\rangle$?

(iii) If $|\psi\rangle$ is normalized then what is the relation between α and β ?

(iv) What do $|0\rangle$ and $|1\rangle$ represent?

(v) What will be the value of $\langle 0|\psi\rangle$?

(5.0 Marks)

2.a. A beam of particles travel from a region where potential $V = 0$ and strike a potential step of height V_0 . if each particle has energy $E (> V_0)$ then give the following:

(i) Schrodinger's equation for the regions where $V = 0$ and $V = V_0$

(ii) Solution to this Schrodinger's equation for both regions

(iii) Boundary conditions

(5.0 Marks)

2.b. An electron is trapped in a 1D infinite potential well of width "a". If the centre of the well is located at the origin then give/show the following:

- (i) Expression for the normalized eigen function for the second excited state,
 - (ii) Expression for the eigen energy for the 4th excited state,
 - (iii) Graph showing probability density for the ground state,
 - (iv) Probability of locating the electron in the 1st excited state between the limits $x = -0.25a$ and $x = 0.25a$,
 - (v) The expression for the magnitude of the momentum in the nth quantum state.
- (5.0 Marks)

2.c. An atom is connected to a rigid support forming a quantum harmonic oscillator.

- (i) If the equilibrium separation of the atom from the support is x_0 , show graphically the behaviour of the potential energy as a function of the distance x from the equilibrium position.
 - (ii) What are the eigen energies of the oscillator?
 - (iii) If the connection of the atom to the support is treated as a spring of constant 0.3N/m , estimate the zero point energy of the oscillator. Take the mass of the atom as $6.7 \times 10^{-27}\text{kg}$.
- (5.0 Marks)

2.d. Compare and contrast the behavior of two identical particles - one trapped in an infinite potential well and the other in a finite potential well. Assume that the wells have the same width.

(6.0 Marks)

2.e. 10^8 electrons strike a potential barrier of height 8eV and width 0.1nm. If each electron has an energy 4eV, estimate the number of electrons that will get reflected of the barrier. (4.0 Marks)

3.a. According to Wiedemann–Franz law , $K/\sigma T = \text{constant}$ (Lorenz number), independent of the metal. The symbols have their usual meaning. Using the quantum free electron theory concepts, arrive at the correct form Lorenz number. Calculate the value and compare it with the experimentally observed value of $L_0 = 2.44 \times 10^{-8} \text{ W}\cdot\Omega/\text{K}^2$. (5.0 Marks)

3.b. A monovalent metal has a free electron concentration $n = 6.0 \times 10^{28} \text{ m}^{-3}$.

(a) Calculate its Fermi energy E_F at 0 K.

(b) Find the corresponding Fermi temperature T_F .

(c) At $T = 300 \text{ K}$, estimate the fraction of electrons that actually participate in conduction (i.e., lie within about $k_B T$ of the Fermi energy).

Assume: $m = 9.11 \times 10^{-31} \text{ kg}$, $h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$,

$k_B = 1.38 \times 10^{-23} \text{ J/K}$ (5.0 Marks)

3.c. Near the bottom of a conduction band, the dispersion relation is experimentally fitted as:

$$E(k) = E_c + \alpha k^2, \alpha = 5.0 \times 10^{-39} \text{ J}\cdot\text{m}^2$$

E_c = lowest point of conduction band, a constant

(a) Determine the effective mass m^* .

(b) Express m^* as a multiple of the free electron mass m . Use :

$$\hbar = 1.055 \times 10^{-34} \text{ J}\cdot\text{s}. \quad (5.0 \text{ Marks})$$

3.d. Explain the concept of molecular field theory due to Weiss. Derive the Curie-Weiss law from the molecular field concept. (5.0 Marks)

3.e. Differentiate between Ferro, Anti-ferro, and Ferri magnetic materials based on their magnetic domains/magnetic moments alignment with some examples. Also, draw a magnetic susceptibility vs temperature for all three, clearly identifying the transition temperatures. (5.0 Marks)

4.a. Consider a hypothetical 3-level and a 4-level laser system. Explain the mechanism of lasing in these lasers. What is the nature of lasing outputs in these lasers and why? Explain. (5.0 Marks)

4.b. Explain the construction and working of a typical homojunction semiconductor laser. How does fabricating a heterojunction leads to charge and photon confinement? what is the advantage? (6.0 Marks)

4.c. Explain various types of polarization mechanisms in a dielectric material. (5.0 Marks)

4.d. A dielectric material contains 8.5×10^{28} atoms per cubic meter. When an external electric field of $E_0 = 1000$ V/m is applied to a parallel-plate capacitor (without dielectric), the surface charge density becomes $\sigma_0 = 8.85 \times 10^{-9}$ C/m².

(a) Calculate the relative permittivity (dielectric constant) ϵ_r if, when the material is inserted, the electric field reduces to 200 V/m.

(b) Calculate the induced polarization P in the material.

(c) Determine the electronic polarizability α_e of individual atoms.

(4.0 Marks)

4.e. A ruby laser emits a line at 694.3 nm. The lifetime of the intermediate metastable state is $\tau = 3.0$ ms. The laser operates with 100 successive pulses per second.

(a) Estimate the natural line width of the ruby laser transition based on the lifetime of the excited state.

(b) Calculate the coherence length of the emitted radiation.

(c) If the laser operates in pulsed mode with pulse duration $t_p = 1.5$ ms, compare the coherence length to the laser cavity length (assuming a 5 cm ruby rod).

(5.0 Marks)